

 AE-AL-2026-DR-01 · REV A · PREPARED FOR REED GOODMAN

Data request — supporting wells

Cards from **additional wells** to calibrate rod-pump diagnostics, and the smaller set of items still outstanding on the well already analysed.

Ordered by how much each changes the answer, not by how easy it is to ask for.

1 Why supporting wells, and what makes one useful

This is the main ask. The physics is done; the diagnostics are not.

Surface-to-downhole card conversion reproduces reference cards to within about 1%. The diagnostic layer that names a pump condition is a different matter: it is currently calibrated on synthetic cards, so it recognises shapes a computer drew rather than shapes a well made. On a real field card it returns answers that do not survive scrutiny. That is a data problem, not an algorithm problem, and it is stated here rather than papered over.

A card on its own contributes almost nothing to fixing that. What makes one valuable is knowing how it turned out — the outcome is the label, and the pulling or workover report is the informative half.

MOST VALUABLE

Card + what was found on the pull	the report is the key half
Before/after pair on one well	same string, known change
Confirmed healthy cards	scarcer than sick ones, and the baseline

ALSO USEFUL

Raw .dyn or timestamped CSV	removes timing error
Repeat cards over months	shows a condition developing
Cards you are unsure about	still useful as unlabelled context



Minimum context per card: rod string, pump size and setting depth, stroke, SPM, and roughly what the well makes.

VALIDATED

2 Still outstanding on the well already analysed

Two questions. Detail and full working are in the engineering report.

Why the load cell leads: the minimum load on the card is 2,412 lb heavier than the entire rod string weighs hanging in air. Friction cannot produce that — on the downstroke it acts upward and lowers polished-rod load, widening a card rather than lifting it. With the string now confirmed as recorded, the load cell is the only remaining explanation.

Why runtime matters more than it looks: a unit on a 50% duty cycle produces half the fluid with a perfectly healthy pump. That is indistinguishable from 50% fillage if all you have is daily volume, so no efficiency figure is meaningful without it.

LOAD CELL

Make and model	needed
Last calibration date	needed
Zeroed with rods hanging or unloaded?	needed

OPERATION

Runtime, hours per day	needed
On a pump-off controller or timer?	needed
Rod string as run	answered – 4,200 ft of 3/4 in ✓



Until the load datum is resolved, absolute loads from this card are not usable. Load differences remain sound, so everything above still stands.

ANALYTICAL

3 Tier 2 — unlocks the downhole card

The piece the thread ended on, and the only way to separate pump-off from gas interference from wear.

Do not over-invest in precision here. Sweeping specific gravity 0.80–0.90 and fluid level 3,000–4,300 ft moves theoretical displacement only from 38.3 to 41.7 bfpd — about $\pm 4\%$. Reasonable estimates are fine; runtime swings the answer far harder.

WELL

Tubing size / ID	needed
Tubing anchored?	needed
Fluid level above pump	acoustic shot if available

FLUID	
Oil gravity, °API	needed
Formation volume factor B_o	needed
Viscosity or bottomhole temperature	needed

4 Tier 3 — unlocks torque, counterbalance and power

The C-66 mentioned in the thread is an Arrow Engine natural-gas engine — the prime mover — not a pumping-unit or gearbox rating, so it gives no torque limit. Without the unit's own designation the torque calculation cannot run at all.

PUMPING UNIT	
Full API designation, e.g. C-228D-200-74	needed
API 11E geometry sheet (A, C, I, K, P, crank radius, phase)	needed
Counterbalance moment and structural imbalance	needed

MOTOR	
Horsepower	needed
Voltage and NEMA class	needed

5 Better data beats more data

A raw controller export — .dyn or timestamped CSV — is worth more than several Tier 2 items. Time read off a position-axis card is inherently uncertain because rod velocity vanishes at both stroke ends: at ± 1.5 in digitising error the peak times carry ± 0.11 to ± 0.15 s. At that uncertainty the apparent peak spacings of 1.24 s and 0.95 s cannot be told apart, nor separated from the predicted 1.03 s. A raw time series removes the error class entirely.

Also useful: several consecutive strokes rather than one, and a card from a different date. Stroke-to-stroke variation distinguishes a steady condition from an intermittent one, and a second date shows whether anything is trending.

6 The trade

What is offered in return, stated plainly.

You send data. We do the analysis and send it back — your wells, worked properly, at no charge. That starts immediately, not at the end of a development programme.

This is an open source effort, not a product built toward a licence fee. The code is already public. Check the arithmetic, hand it to your own engineer, or ignore us entirely and still use it. There is nothing to get locked into and nothing that can be taken away later.

Early contributors are supported for life — as stated, not an introductory rate that changes once the work matures. And it does not stop at dynacards: rod string design, unit sizing, gas interference, production troubleshooting, anything across your operations.



Well identifiers are kept out of anything public. The existing validation set is anonymised the same way.

VALIDATED

7 What comes back

Forward this brief to anyone it might suit — the same terms apply.

Every well sent is analysed and returned: card metrics, the calculated downhole card where the inputs allow it, and an honest statement of what each result can and cannot support. Same format as the report linked below.

Approximately three months from a workable labelled set, a diagnostics package carrying a measured accuracy figure — not a marketing one. Contributors' wells run through it first.



Until a real-card accuracy figure exists, any diagnosis from this system is a hypothesis to check against the well, not a finding. The downhole card and geometric measurements are trustworthy today; the labels placed on them are not yet.

PENDING

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Analysis of the well already reviewed: [engineering report AE-AL-2026-001](#). Prepared by AceEngineer Artificial Lift.